

Unit-IV Tutorial

① Let P be a stochastic matrix

$$P = \begin{bmatrix} 0 & 0.3 & 0.7 \\ 1 & 0 & 0 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

$$\text{Now, } P^2 = P \cdot P = \begin{bmatrix} 0 & 0.3 & 0.7 \\ 1 & 0 & 0 \\ 0.2 & 0.2 & 0.6 \end{bmatrix} \begin{bmatrix} 0 & 0.3 & 0.7 \\ 1 & 0 & 0 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$

$$= \begin{bmatrix} 0.44 & 0.14 & 0.42 \\ 0 & 0.3 & 0.7 \\ 0.32 & 0.18 & 0.5 \end{bmatrix}$$

$$P^3 = P^2 \cdot P = \begin{bmatrix} 0.44 & 0.14 & 0.42 \\ 0 & 0.3 & 0.7 \\ 0.32 & 0.18 & 0.5 \end{bmatrix} \begin{bmatrix} 0 & 0.3 & 0.7 \\ 1 & 0 & 0 \\ 0.2 & 0.2 & 0.6 \end{bmatrix} = \begin{bmatrix} 0.226 & 0.56 & 0.114 \\ 0.44 & 0.42 & 0.196 \\ 0.28 & 0.52 & 0.2 \end{bmatrix}$$

Here all elements > 0 , $n=3$, $\therefore P$ is regular.

② Given, $P = \begin{bmatrix} 0.2 & 0.3 & 0.5 \\ 0.1 & 0.2 & 0.7 \\ 0.6 & 0.3 & 0.1 \end{bmatrix}$

Also, $P^{(0)} = \begin{bmatrix} 0.4 & 0.3 & 0.3 \end{bmatrix}$

$$P^{(1)} = P^{(0)} \cdot P = \begin{bmatrix} 0.4 & 0.3 & 0.3 \end{bmatrix} \begin{bmatrix} 0.2 & 0.3 & 0.5 \\ 0.1 & 0.2 & 0.7 \\ 0.6 & 0.3 & 0.1 \end{bmatrix} = \begin{bmatrix} 0.29 & 0.27 & 0.44 \end{bmatrix}$$

- ③ The travel pattern is a markov chain, with state space = (train, car)

$$\therefore P = \begin{matrix} & \begin{matrix} T & C \end{matrix} \\ \begin{matrix} T \\ C \end{matrix} & \begin{bmatrix} 0 & 1 \\ 1/2 & 1/2 \end{bmatrix} \end{matrix}$$

let $V = [x \ y]$

then $VP = V$

$$\Rightarrow [x \ y] \begin{bmatrix} 0 & 1 \\ 1/2 & 1/2 \end{bmatrix} = [x \ y]$$

$$\Rightarrow \frac{x}{2} = x ; \quad x + \frac{y}{2} = y \quad \text{Also } x_1 + x_2 = 1$$

on solving we get,

$$x = 1/3, \quad y = 2/3$$

$$\therefore V = [1/3 \ 2/3] ; \text{ The prob. of the man goes by train} = 1/3$$

- ④ let $X(t)$ be a poisson process with mean $\mu = \lambda t$
 $\lambda = 2$, $\mu = 2t$, $P[X(t) = n] = \frac{e^{-\lambda t} (\lambda t)^n}{n!}$

$$i) P[X(1/2) = 1] = \frac{e^{-2t} (2t)^n}{n!} = \frac{e^{-2(1/2)} (2 \cdot 1/2)^1}{1!} = 0.3679$$

$$ii) P[X(2) = 3] = \frac{e^{-2(2)} \cdot (2(2))^3}{3!} = 0.1954$$

⑤ Let $X(t)$ be the number of calls arrive at a board in the time interval of t .

i) Given: Mean rate $\lambda = 2/\text{minute}$

Time interval $t = 5/\text{minute}$

No. of arrivals, $n = 6$

$$\therefore P[X(t) = n] = \frac{e^{-\lambda t} (\lambda t)^n}{n!}$$

$$P[X(5) = 0] = \frac{e^{-2 \times 5} [2 \times 5]^0}{0!} = e^{-10} = 4.54 \times 10^{-5}$$

ii) Given: Mean rate, $\lambda = 2/\text{minute}$

Time interval, $t = \frac{1}{2}/\text{minute}$

No. of arrivals, $n = \text{More than one call}$

$$P[X(\frac{1}{2}) > 1] = 1 - P[X(\frac{1}{2}) \leq 1]$$

$$= 1 - [P[X(\frac{1}{2}) = 0] + P[X(\frac{1}{2}) = 1]]$$

$$= 1 - \left(\frac{e^{-1} (1)^0}{0!} + \frac{e^{-1} (1)^1}{1!} \right) = 1 - 2e^{-1} = 0.2642$$